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Liquid ejection apparatus

Abstract

A liquid ejection apparatus includes a liquid ejection section configured to print by discharging a liquid from a nozzle, a liquid container having an accommodation chamber that stores liquid and an injection port that communicates with the accommodation chamber and into which the liquid is configured to be injectable from the outside, a supply flow path communicating between the liquid ejection section and the liquid container, and an opening and closing mechanism. The opening and closing mechanism that is configured to switch such that an open state in which the supply flow path is opened and a closed state in which the supply flow path is closed when power is turned on and not to switch when power is turned off.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-092784, filed Jun. 8, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.
BACKGROUND
1. Technical Field
(2) The present disclosure relates to a liquid ejection apparatus including a liquid ejection section that ejects liquid.
2. Related Art
(3) JP-A-2020-168842 discloses a liquid ejection apparatus including a liquid ejection section for ejecting liquid. The liquid ejection apparatus includes a liquid container for containing liquid, a liquid ejection section for ejecting the liquid, and a supply flow path for supplying the liquid from the liquid container to the liquid ejection section. In addition, the liquid ejection apparatus includes an opening and closing mechanism having a manually operated lever capable of opening and

closing the supply flow path. A user, a repairman, or the like can operate to close the supply flow path when necessary by operating the lever of the opening and closing mechanism. For example, when the liquid ejection apparatus is transported, the lever is operated to close the supply flow path.
(4) However, in the liquid ejection apparatus described in JP-A-2020-168842, when a user, a transporter, a repairman, or the like intentionally or erroneously operates a manual operation section such as a lever with respect to the liquid ejection apparatus in a power-off state, the liquid ejection apparatus is transported in a state in which the supply flow path is opened. In this case, there is a risk that liquid such as ink may leak from the liquid ejection section during transportation. As described above, in the related liquid ejection apparatus, there is a problem that a malfunction occurs because a manual operation can be performed intentionally or erroneously with respect to the liquid ejection apparatus in the power-off state.
SUMMARY
(5) A liquid ejection apparatus that solves the above problem includes a liquid ejection section configured to print by ejecting a liquid from a nozzle, a liquid container including an accommodation chamber configured to store liquid and an injection port that communicates with the accommodation chamber and that is configured to be injected with liquid from outside, a supply flow path communicating between the liquid ejection section and the liquid container, and an opening and closing mechanism configured to, when power is turned on, enable, and, when power is turned off, disable switching between an open state, in which the supply flow path is open, and a closed state, in which the supply flow path is closed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view showing a liquid ejection apparatus according to an embodiment.
(2) FIG. **2** is a perspective view of the liquid ejection apparatus showing how the liquid is injected into the liquid container.
(3) FIG. **3** is a front view of the liquid ejection apparatus, partially broken away so that the liquid container can be seen.
(4) FIG. **4** is a perspective view showing a liquid container, an opening and closing mechanism, a liquid ejection section, a carriage, and the like.
(5) FIG. **5** is a schematic diagram illustrating a liquid container, a supply flow path, an atmosphere communication flow path, an opening and closing mechanism, a liquid ejection section, and the like.
(6) FIG. **6** is a perspective view showing the opening and closing mechanism.
(7) FIG. **7** is an exploded perspective view of the opening and closing mechanism.
(8) FIG. **8** is a cross-sectional view showing a state in which the opening and closing mechanism opens the flow path.
(9) FIG. **9** is a cross-sectional view showing a state in which the opening and closing mechanism closes the flow path.
(10) FIG. **10** is a schematic diagram showing a configuration of a liquid container, a flow path, an opening and closing mechanism, and a liquid ejection section.
(11) FIG. **11** is a diagram showing a setting screen.
(12) FIG. **12** is a block diagram showing an electrical configuration of the liquid ejection apparatus.
(13) FIG. **13** is a table showing four switching positions of the opening and closing mechanism.
(14) FIG. **14** is a flowchart showing control contents when the transport mode is turned on and off.
(15) FIG. **15** is a flowchart showing control contents when the replacement of the liquid ejection section is performed.

(16) FIG. **16** is a flowchart showing control contents when the cover is opened and closed.
(17) FIG. **17** is a schematic diagram showing a liquid container, a supply flow path, an atmosphere communication flow path, an opening and closing mechanism, a liquid ejection section, and the like in a modification.
(18) FIG. **18** is a schematic diagram showing a liquid container, a supply flow path, an atmosphere communication flow path, an opening and closing mechanism, a liquid ejection section, and the like in a modification different from FIG. **17**.

DESCRIPTION OF EMBODIMENTS

(19) Hereinafter, a liquid ejection apparatus according to an embodiment will be described with reference to the drawings.
(20) As shown in FIG. **1**, a liquid ejection apparatus **11** includes a printing section **12** for printing on a medium M, such as paper, and a scanner **13** for reading an original (not shown). The scanner **13** is disposed above the printing section **12**. The liquid ejection apparatus **11** includes a device main body **14** constituting the printing section **12**. The device main body **14** has an opening opened upward in the upper portion thereof. The scanner **13** functions as an openable and closable cover **13**A that covers the opening of the device main body **14**.
(21) In FIG. **1**, assuming that the liquid ejection apparatus **11** is placed on a horizontal plane, a direction along a gravity direction is defined as a vertical direction Z, and directions along the horizontal plane are defined as a width direction X and a depth direction Y. That is, the width direction X, the depth direction Y, and the vertical direction Z intersect with each other (preferably orthogonally). In addition, one end side in the depth direction Y may be referred to as a front surface side or a front side, the other end side opposite to the one end side may be referred to as a back surface side or a rear side, one end side in the width direction X as viewed from the front surface side may be referred to as a right side, and the other end side may be referred to as a left side. In addition, since the width direction X is also a main scanning direction in which a liquid ejection section **31** mounted on a carriage **30** (to be described later) moves at the time of printing, the width direction X is also referred to as the main scanning direction X. In this case, a direction parallel to and opposite to the depth direction Y in which the medium M is transported at the printing position, which faces the liquid ejection section **31**, is also referred to as a transport direction −Y or a sub-scanning direction −Y.
(22) As shown in FIG. **1**, the device main body **14** includes the liquid ejection section **31**, the carriage **30**, a transport mechanism **70** (refer to FIG. **12**), liquid storage units **20**, supply flow paths **24**, an opening and closing mechanism **40**, and a control section **90** (refer to FIG. **12**).
(23) The device main body **14** includes a housing **14**A, which is the exterior case of the device main body **14**. The liquid ejection section **31**, the carriage **30**, the supply flow paths **24**, the opening and closing mechanism **40**, the transport mechanism **70**, and the control section **90** are disposed in the housing **14**A. The liquid storage units **20** are disposed on both sides in the width direction X in the front portion of the housing **14**A.
(24) Each liquid storage unit **20** includes a container case **21** and a cover **22** that covers an opening of the container case **21**. Each container case **21** accommodates liquid containers **23** (see FIGS. **2** and **3**) for accommodating a liquid. Each cover **22** is provided so as to be openable and closable with respect to the container case **21**. More specifically, each cover **22** is configured to be displaceable between a cover position shown in FIG. **1** for covering the liquid containers **23** and an exposed position shown in FIG. **2** for exposing the liquid containers **23**. The liquid ejection apparatus **11** includes, for example, cover sensors **71** (refer to FIG. **12**) as an example of displacement detection units that detect displacement of each cover **22**. For example, a user or the like displaces the cover **22** from the cover position to the exposed position when refilling the liquid to the liquid containers **23**. The displacement of the cover **22** is detected by the cover sensor **71**. The control section **90** can detect the displacement of the cover **22** by the detection signal of the cover sensor **71**.

(25) As shown in FIG. **1**, an operation panel **17** including an operation section **15**, such as buttons operated to give various instructions to the liquid ejection apparatus **11**, and a display section **16** that displays information such as a menu, is provided on the front side of the liquid ejection apparatus **11**. The display section **16** is configured by, for example, a liquid crystal display panel. When the display section **16** is a touch panel, a touch operation function of the display section **16** may constitute a part of the operation section **15**. The operation section **15** includes a power operation section **15**A that is operated when turning on and off the power of the liquid ejection apparatus **11**.
(26) As shown in FIG. **1**, a medium accommodation section **18** that accommodates the medium M is attachably and detachably inserted into a lower portion of the device main body **14**. The medium accommodation section **18** is, for example, a cassette capable of storing the medium M. The plurality of media M stored in the medium accommodation section **18** are fed one sheet at a time along a predetermined transport path. The medium M is transported in the transport direction −Y through a transport region FA (refer to FIG. **4**) that has a width dimension that is slightly narrower than the width dimension of the medium accommodation section **18** in the width direction X. In a process in which the medium M is transported in the transport region FA, the liquid is ejected to the medium M by the liquid ejection section **31**. The medium accommodation section **18** is not limited to a single level, and may be provided as a plurality of levels. The supply source of the medium M is not limited to the medium accommodation section **18** such as a cassette, and it may be a medium mounting section including a tray on which one or a plurality of media M can be placed.
(27) The scanner **13** is configured to be rotatable with respect to the device main body **14** with a rear surface side of the device main body **14** as a rotation fulcrum. The scanner **13** is rotatable between a closed posture (refer to FIG. **1**) and an open posture (refer to FIG. **2**) with respect to the device main body **14**. The scanner **13** functioning as the cover **13**A can open and close the device main body **14** between a cover position at which the scanner **13** covers the inside of the device main body **14** and an exposed position at which the inside of the device main body **14** is exposed. The liquid ejection apparatus **11** may not include the scanner **13**. Instead of the scanner **13**, a simple cover **13**A that covers the opening of the device main body **14** from the upper surface may be provided.
(28) As shown in FIG. **2**, the scanner **13** is attached via a rotation mechanism **13**B such as a hinge provided on the back surface side. The scanner **13** can open and close with respect to the printing section **12**, and rotates between a closed position shown in FIG. **1** and an open position shown in FIG. **2**.
(29) As shown in FIG. **1**, in a state in which the cover **13**A is closed, a part of the cover **13**A covers a part of the cover **22** in the closed state. Therefore, in order to open the cover **22**, it is necessary to first open the cover **13**A. When the cover **13**A is rotated to the open position, the cover **22** of the liquid storage units **20** can open and close. The cover **22** is configured to be displaced between a cover position at which the liquid containers **23** are covered and an exposed position at which the liquid containers **23** are exposed. Note that the cover **22** may be omitted, and the cover **13**A may be configured to be displaced between a cover position at which the cover covers the liquid containers **23** and an exposed position at which the liquid containers **23** are exposed.
(30) The liquid containers **23** are each provided with an injection section **27** having an injection port **27**A at its upper part and a lid member **28** capable of stoppering the injection port **27**A at a closed position. The lid member **28** is configured to be displaced between the closed position in which the injection section **27** is stoppered and an open position in which the injection section **27** is not stoppered. When the cover **22** is rotated to the exposed position (open position), the lid member **28** can open and close. When the liquid is supplied to a liquid container **23**, as shown in FIG. **2**, the scanner **13**, the cover **22**, and the lid member **28** are sequentially rotated to the open position, and a liquid bottle **75** filled with the liquid is coupled to the injection section **27** of the liquid container **23** in an inverted posture. The liquid in the liquid bottle **75** is injected into the liquid container **23** via

the injection section **27**.

(31) Two liquid storage units **20** in this example are disposed on both sides of the operation panel **17** in the width direction X. Here, the operation panel **17** is disposed in substantially the same region as the transport region FA of the medium M in the width direction X. For this reason, the plurality of liquid containers **23** are separately disposed on both sides of the transport region FA of the medium M.

(32) As shown in FIGS. **2** and **3**, one liquid storage unit **20** contains three liquid storage sections **23**, and the other liquid storage unit **20** contains one liquid container **23**. Three liquid containers **23** in the one liquid storage unit **20** contain, for example, color ink used at the time of color printing, as an example of liquid. For example, inks of three colors of cyan, magenta, and yellow are respectively stored in the three liquid containers **23**. For example, black ink used at the time of monochrome printing or the like is stored as an example of liquid in one liquid container **23** in the other liquid storage unit **20**. The ink may be either a pigment type or a dye type.

(33) In the liquid ejection apparatus **11**, a first liquid container **23**A and a second liquid container **23**B having different liquid amount capacities are attached side by side in the widthwise direction X. In the present embodiment, one first liquid container **23**A for black having a large capacity is provided on the left side with respect to the operation panel **17**, and the three second liquid containers **23**B, for color, having a smaller capacity than the first liquid container **23**A, are provided on the right side with respect to the operation panel **17**. Note that the plurality of second liquid containers **23**B have the same configuration, and the same reference numerals are given to configurations that are common between the first liquid container **23**A and the second liquid containers **23**B, and duplicate description thereof will be omitted.

(34) The volume of the first liquid container **23**A accommodating black ink is larger than the volume of the second liquid container **23**B accommodating color ink. This is because the amount of liquid consumption is greater for black ink than for the color ink. In the example shown in FIGS. **2** to **4**, one first liquid container **23**A for containing black ink and the plurality (for example, three) of the second liquid containers **23**B for containing color ink are disposed on the left and right sides separately, but the way to divide and arrange the plurality of liquid containers **23** on the left and right can be appropriately changed.

(35) The container case **21** constituting the liquid storage unit **20** is provided with window sections **21**A at positions corresponding to the liquid containers **23**. A user can visually check the residual amount of the liquid in the liquid containers **23** through the window sections **21**A.

(36) As shown in FIG. **3**, each liquid container **23** has an accommodation chamber **26** holding liquid, and the injection port **27**A that communicating with the accommodation chamber **26** and into which liquid can be injected from the outside. Note that the first liquid container **23**A and the second liquid containers **23**B have basically the same configuration except that their widths are slightly different due to their different volumes.

(37) Each liquid container **23** shown in FIG. **3** is at least partially made of transparent or translucent resin, and the liquid level of the liquid accommodated in the accommodation chamber **26** can be visually checked from the outside. The window sections **21**A of the container cases **21** (refer to FIG. **1**) function as a viewer surface through which liquid in the liquid containers **23** can be visually checked from outside. The window section **21**A is provided with a lower limit mark LL for gauging the lower limit when the liquid needs to be replenished in the accommodation chamber **26**, and an upper limit mark UL for gauging the upper limit of the replenishing amount of the liquid. Note that the viewer surface is provided along the vertical direction Z in the use state of the first liquid container **23**A.

(38) Further, the upper limit mark UL may be provided on the window sections **21**A instead of the liquid containers **23**, for example, the window section **21**A may be formed as a light transmitting portion that transmits light by a transparent or translucent member. It is also possible not to provide the upper limit mark UL.

(39) Further, the liquid ejection apparatus **11** shown in FIG. **1** includes the liquid ejection section **31** in the housing **14**A, for performing printing by ejecting liquid from nozzles **33** (refer to FIG. **10**). The liquid ejection apparatus **11** includes the supply flow paths **24** having tubes or the like for supplying the liquid contained in the liquid containers **23** to the liquid ejection section **31**. The liquid ejection apparatus **11** includes the carriage **30** on which the liquid ejection section **31** is mounted and which can reciprocate along the main scanning direction X. The liquid ejection section **31** includes a liquid ejection head **32** (refer to FIGS. **4** and **10**) that ejects liquid from the nozzles **33**. The liquid ejection section **31** ejects liquid toward the medium M from the nozzles **33** of the liquid ejection head **32** while moving in the main scanning direction X, thereby printing characters, images, or the like on the medium M.
(40) As shown in FIGS. **1** and **3**, the liquid ejection apparatus **11** includes the supply flow paths **24** that bring the liquid ejection section **31** and the liquid containers **23** into communication with each other. The supply flow paths **24** are, for example, tubes. The liquid ejection apparatus **11** includes the opening and closing mechanism **40** configured to open and close the supply flow paths **24**.
(41) Further, each liquid container **23** has an atmosphere communication flow path **25** capable of bringing the inside of the accommodation chamber **26** into communication with atmosphere. The opening and closing mechanism **40** is configured to open and close the atmosphere communication flow paths **25**. The supply flow paths **24** and the atmosphere communication flow paths **25** are coupled to the opening and closing mechanism **40**. Through the opening and closing mechanism **40**, the supply flow paths **24** are coupled to the liquid ejection section **31** by a predetermined flow path route that turns back with a curve along the way in the width direction X. The supply flow paths **24** have a configuration of four tubes for supplying four colors of ink, bundled horizontally adjacent to each other.
(42) In addition, the opening and closing mechanism **40** is configured such that an open state, in which the supply flow paths **24** are open, and a closed state, in which the supply flow paths **24** are closed, can be switched between when the power is on and cannot be switched between when the power is off. The liquid ejection apparatus **11** is configured to enable selection of a first mode in which, when power is turned off, power is turned off with the opening and closing mechanism **40** in the open state and a second mode in which, when power is turned off, power is turned off with the opening and closing mechanism **40** in the closed state.
(43) As shown in FIGS. **1** and **3**, the liquid ejection apparatus **11** includes the carriage **30** on which the liquid ejection section **31** is mounted and which is movable in the main scanning direction X.
(44) Internal Configuration of the Printing Section **12**
(45) Next, internal configuration of the printing section **12** will be described with reference to FIG. **4**. As shown in FIG. **4**, the carriage **30** can reciprocate in the main scanning direction X by a driving force applied from a carriage motor **36**. The liquid ejection section **31** is mounted on the carriage **30**.
(46) An endless timing belt **38** wound around a pair of pulleys **37** extends along the main scanning direction X on the back side of the movement path of the carriage **30**. A driving pulley of the pair of pulleys **37** is fixed to a rotation shaft (not illustrated) of the carriage motor **36**. At least a part of the timing belt **38** is fixed to a back side end portion of the carriage **30**. When the carriage motor **36** is driven in forward and reverse directions, the timing belt **38** rotates in the same direction as the rotation direction of the carriage motor **36**, and causes the carriage **30** to reciprocate in the main scanning direction X.
(47) The carriage **30** shown in FIG. **4** reciprocates along a guide shaft (not shown) extending in the main scanning direction X. The carriage **30** waits at a standby position HP (home position) during non-printing. A maintenance unit **50** for performing maintenance of the liquid ejection section **31** is disposed immediately below the carriage **30** that moved to the standby position HP. The maintenance unit **50** includes a cap **51** as an example of a closed space formation section that forms a closed space in which the nozzles **33** open at a standby position HP in which the liquid ejection

section **31** can wait. The cap **51** can abut on the liquid ejection section **31**, for example, so as to surround the nozzles **33**. In addition, the maintenance unit **50** includes a pump **53** that reduces the pressure of a closed space that is formed when the cap **51** abuts on a nozzle surface **32**A (refer to FIG. **10**) in which the nozzles **33** of the liquid ejection head **32** are opened. When the closed space formed by the cap **51** abutting on the nozzle surface **32**A is decompressed, cleaning of the nozzles **33** for discharging unnecessary ink and air bubbles in the nozzles **33** is performed.
(48) In the liquid ejection apparatus **11** of the present embodiment, a liquid receiving section (not illustrated) is provided directly below the carriage **30** when the carriage **30** moves to a non-printing region that is located on both sides of a transport region FA (printing region) in the main scanning direction X. The liquid receiving section receives ink discharged from the nozzles **33** by dummy ejection, which is a type of maintenance. Dummy ejections are for driving the piezoelectric element to discharge ink that is not used for printing from the nozzles **33** and eliminate thickening of ink in the nozzle **33**. The cap **51** may also serve as the liquid receiving section.
(49) Further, the liquid ejection apparatus **11** has the transport mechanism **70** (refer to FIG. **12**) including a plurality of rollers for transporting the sheet-like medium M. The transport mechanism **70** transports the medium M in a transport direction −Y that intersects with the main scanning direction X, which is the moving direction of the carriage **30** (liquid ejection section **31**). A support base (platen) (not shown) is provided so as to face the liquid ejection head **32** below the range in which the liquid ejection section **31** moves. The support base supports the medium M transported by the transport mechanism **70**. The liquid ejection section **31** prints an image or the like on the medium M by ejecting liquid such as ink onto a portion of the medium M that is transported in the transport region FA and that is supported by the support base.
(50) With respect to the positional relationship between the liquid containers **23** and the liquid ejection section **31**, the liquid surface in the liquid containers **23** is provided at a position lower than the nozzle **33** of the liquid ejection section **31** by a predetermined height in the vertical direction Z. That is, a negative pressure due to a water head difference by a predetermined height is applied to the nozzles **33**.
(51) The liquid ejection section **31** shown in FIG. **4** is provided so as to be attachable to and detachable from the carriage **30**. One end of each supply flow path **24** is provided so as to be attachable to and detachable from the liquid ejection section **31**. The liquid ejection section **31** includes a joint section **35** coupled to the supply flow paths **24**.
(52) The liquid ejection section **31** of present embodiment is configured to be replaceable with respect to the carriage **30**. The carriage **30** is provided with an attach and detach operation section **34** rotatably supported around a shaft. A user can attach and detach the liquid ejection section **31** to and from the carriage **30** by operating the attach and detach operation section **34**. The attachment and detachment work of the liquid ejection section **31** is performed at an exchange position EP illustrated in FIG. **4** in a state in which the cover **13**A including the scanner **13** is opened. The exchange position EP is a position different from the standby position HP in the main scanning direction X. In response to a command to replace the liquid ejection section **31**, the carriage **30** moves from the standby position HP to the exchange position EP.
(53) As shown in FIG. **4**, the joint section **35** is a member that couples the supply flow paths **24** and the flow path of the liquid ejection section **31**. The joint section **35** is provided, for example, in the attach and detach operation section **34**. When the user operates the attach and detach operation section **34** in an opening direction in order to detach the liquid ejection section **31** from the carriage **30**, the joint section **35** is detached from the liquid ejection section **31**. By this, the supply flow paths **24** and the liquid ejection section **31** are separated from each other. Then, the liquid ejection section **31** can be detached from the carriage **30** in a state of being separated from the supply flow paths **24**. In addition, when the user attaches the liquid ejection section **31** to the carriage **30**, the attach and detach operation section **34** is closed, and a pressing unit (not illustrated) is pressed. Accordingly, the connection between the joint section **35** and the liquid ejection section **31** is

realized. Due to the connection between the joint section **35** and the liquid ejection section **31**, the supply flow paths **24** and the flow path of the liquid ejection section **31** communicate with each other again, and the liquid can be supplied to the liquid ejection head **32**.
(54) About the Opening and Closing Mechanism **40**
(55) Next, a detailed configuration of the opening and closing mechanism **40** will be described with reference to FIGS. **4** and **5**. As shown in FIG. **5**, the opening and closing mechanism **40** includes a first opening and closing section **41** and a second opening and closing section **42**. As shown in FIG. **4**, the opening and closing mechanism **40** includes a drive mechanism **43** that outputs a driving force for opening and closing the flow paths **24** and **25**. The drive mechanism **43** includes a motor **43**M as a drive section. The first opening and closing section **41** shown in FIG. **5** is configured to open and close the supply flow paths **24**. Further, the second opening and closing section **42** is configured to open and close the atmosphere communication flow paths **25**. The control section **90** drives and controls the motor **43**M to control the opening and closing state of the first opening and closing section **41** and the second opening and closing section **42**.
(56) A state in which the opening and closing mechanism **40** closes only the supply flow paths **24** and a state in which the opening and closing mechanism **40** closes only the atmosphere communication flow paths **25** can be selected. When the first opening and closing section **41** is closed and the second opening and closing section **42** is open, then the opening and closing mechanism **40** closes only the supply flow paths **24** out of the supply flow paths **24** and the atmosphere communication flow paths **25**. When the first opening and closing section **41** is open and the second opening and closing section **42** is closed, then the opening and closing mechanism **40** closes only the atmosphere communication flow paths **25** out of the supply flow paths **24** and the atmosphere communication flow paths **25**. In the opening and closing mechanism **40** of the present embodiment, there are four combinations of open and closed states of the supply flow paths **24** and the atmosphere communication flow paths **25**. The combination of the open and closed states may be two or three.
(57) Detailed Configuration of the Opening and Closing Mechanism **40**
(58) Next, detailed configuration of the opening and closing mechanism **40** will be described with reference to FIGS. **6** and **7**. FIG. **6** is a perspective view of the opening and closing mechanism **40**. FIG. **7** is an exploded perspective view of the opening and closing mechanism **40**.
(59) The liquid ejection apparatus **11** includes the motor **43**M configured to drive the opening and closing mechanism **40**.
(60) As illustrated in FIG. **7**, the opening and closing mechanism **40** includes, as the first opening and closing section **41** capable of opening and closing the supply flow paths **24**, a first pressing member **46** capable of pressing the supply flow paths **24** and a first cam **44** configured to switch a pressing state of the first pressing member **46** by driving of the motor **43**M. Further, the opening and closing mechanism **40** includes, as the second opening and closing section **42** capable of opening and closing the atmosphere communication flow paths **25**, a second pressing member **47** capable of pressing the atmosphere communication flow paths **25**, and a second cam **45** configured to switch a pressing state of the second pressing member **47** by driving the motor **43**M. The drive mechanism **43** may include a drive transmission section **43**A for transmitting the drive force of the motor **43**M. The drive transmission section **43**A includes a rotation shaft, a gear, and the like. The drive transmission section **43**A changes the speed of the drive force of the motor **43**M and transmits it to the cams **44** and **45**.
(61) As shown in FIGS. **6** and **7**, the opening and closing mechanism **40** includes the cams **44** and **45**, the pressing members **46** and **47**, and a flow path holding case **49** that holds the flow paths **24** and **25** in a sandwiched state. The flow path holding case **49** is an exterior case of the opening and closing mechanism **40**. The flow path holding case **49** includes a flow path support section **49**A and a flow path covering section **49**B.
(62) The flow path support section **49**A is provided with a plurality of recesses **49**E extending in a

direction intersecting the axial direction of the cams **44** and **45**. The supply flow paths **24** and the atmosphere communication flow paths **25** are inserted into the plurality of recesses **49**E. The first pressing member **46** is disposed on the upper side of the four supply flow paths **24**, and the second pressing member **47** is disposed above the one atmosphere communication flow path **25**. Further, the first cam **44** is disposed above the first pressing member **46**, and the second cam **45** is disposed above the second pressing member **47**. The flow paths **24** and **25**, the pressing members **46** and **47**, and the cams **44** and **45** are arranged in this order from the flow path support section **49**A side, in between the flow path support section **49**A and the flow path covering section **49**B.
(63) The flow path covering section **49**B has a plurality of engaging sections **49**C. Further, the flow path support section **49**A has a plurality of engaged sections **49**D at positions corresponding to the plurality of engaging sections **49**C. By locking the plurality of engaging sections **49**C with the plurality of engaged sections **49**D, the flow path support section **49**A and the flow path covering section **49**B are fixed together as the single flow path holding case **49**. By this, the opening and closing mechanism **40** shown in FIG. **6** is constructed.
(64) The first opening and closing section **41** includes the first cam **44** and the first pressing member **46**. When the first cam **44** is positioned at the rotation angle of the closed position, the first opening and closing section **41** closes the supply flow paths **24** by the first cam **44** pushing the first pressing member **46**. The second opening and closing section **42** includes the second cam **45** and the second pressing member **47**. When the second cam **45** is positioned at the rotation angle of the closed position, the second opening and closing section **42** closes the atmosphere communication flow paths **25** by the second cam **45** pushing the second pressing member **47**.
(65) As shown in FIG. **7**, the cams **44** and **45** are arranged coaxially, and one axial end portion thereof is coupled to the motor **43**M via the drive transmission section **43**A. The cams **44** and **45** are integrally formed of, for example, resin. Further, the pressing members **46** and **47** and the flow path holding case **49** are, for example, also molded products of resin. The atmosphere communication flow path **25** is divided into three parts including a first flow path section **25**A coupled to the first liquid container **23**A and a second flow path section **25**B coupled to the second liquid container **23**B. An opening end of the flow path section excluding the first flow path section **25**A and the second flow path section **25**B is an air opening port **25**C.
(66) Operation of the Opening and Closing Mechanism **40**
(67) Next, the operation of the opening and closing mechanism **40** will be described with reference to FIGS. **8** and **9**. FIG. **8** shows a state when the opening and closing mechanism **40** is in the open position. FIG. **9** shows a state in which the opening and closing mechanism **40** is in the closed position. FIGS. **8** and **9** are sectional views of the opening and closing mechanism **40** as viewed in the axial direction of the cams **44** and **45**, and show an open and closed state of a portion of the supply flow paths **24**. FIG. **8** shows an open state of the supply flow paths **24**, and FIG. **9** shows a closed state of the supply flow paths **24**. In FIGS. **8** and **9**, the flow path covering section **49**B of the flow path holding case **49** is not shown.
(68) As shown in FIGS. **8** and **9**, the cams **44** and **45** are formed in such a shape that the diameter lengths thereof with respect to the rotation center C**1** vary depending on the position in the circumferential direction. A shape of an outer peripheral surface forming a cam surface of the cam **44** is, for example, an elliptical shape. The control section **90** rotationally drives the motor **43**M to rotate the cams **44** and **45**, for example, in the clockwise direction in FIGS. **8** and **9**. As shown in FIGS. **8** and **9**, when the cam **44** (**45**) rotates, the position where the outer peripheral surface of the cam **44** (**45**) abuts on the pressing member **46** (**47**) changes, and the pressing members **46** (**47**) are displaced in the direction in which they can approach and separate from the flow path **24** (**25**).
(69) As shown in FIG. **8**, when the cam **44** is in a posture in which the long diameter direction of the cam **44** and the supply flow paths **24** are substantially parallel to each other, the pressing member **46** (**47**) is displaced in a direction away from the flow path **24** (**25**). Therefore, the amount of displacement, in the direction in which the pressing member **46** (**47**) presses the supply flow

paths **24**, is reduced, and the flow path **24** (**25**) is opened.

(70) As shown in FIG. **9**, when the cam **44** enters a posture in which the long diameter direction of the cam **44** and the supply flow path **24** are substantially perpendicular to each other, the pressing member **46** (**47**) is displaced in a direction to approach the flow path **24** (**25**). Therefore, the amount of displacement increases in the direction in which the pressing member **46** (**47**) presses the supply flow path **24**, and the flow paths **24** (**25**) are closed. In this manner, the opening and closing mechanism **40** opens and closes the four supply flow paths **24**.

(71) On the other hand, the outer peripheral surface shape of the second cam **45** may be different from the outer peripheral surface shape of the first cam **44**. Since the cams **44** and **45**, in the present embodiment, coaxially rotate about the rotation center C**1**, the second cam **45** may be formed to have a predetermined outer peripheral surface shape so that a desired combination of open and closed states (for example, the combination shown in FIG. **13**) can be realized even when the cam coaxially rotates. For example, the cams **44** and **45** have a short diameter portion and a long diameter portion. The first cam **44** has an ellipsoidal outer peripheral surface shape in which a portion that abuts the first pressing member **46** changes in one rotation in the order of a short diameter portion at 0 degrees, a long diameter portion at 90 degrees, a short diameter portion at 180 degrees, and a long diameter portion at 270 degrees. On the other hand, the second cam **45** has an outer peripheral surface shape in which a portion that abuts on the second pressing member **47** changes in one rotation in the order of a short diameter portion at 0 degrees, a short diameter portion at 90 degrees, a long diameter portion at 180 degrees, and a long diameter portion at 270 degrees.

(72) For example, when the liquid ejection apparatus **11** is transported, if the supply flow paths **24** are closed by the opening and closing mechanism **40**, ink is less likely to leak from the nozzles **33** of the liquid ejection section **31**. When the liquid ejection apparatus **11** is transported, vibration or impact acts on the liquid in the liquid containers **23** or the supply flow paths **24**. As a result, pressure acts on the liquid in the nozzles **33** of the liquid ejection section **31**, and there is a risk that the liquid such as ink leaks from the nozzles **33**.

(73) If the supply flow paths **24** is closed by the opening and closing mechanism **40** before the liquid ejection apparatus **11** is transported, it is possible to suppress leakage of liquid such as ink from the nozzles **33** even if there is a pressure fluctuation acting on the liquid in the liquid ejection section **31** when the liquid ejection apparatus **11** is transported.

(74) In the present embodiment, a user, a transporter, or a repairman puts the liquid ejection apparatus **11** into a transport mode during power-on, and then powers off the liquid ejection apparatus **11**. As a result, the motor **43**M is driven before the power is turned off, so that the first opening and closing section **41** is displaced in the closed position and the supply flow paths **24** are closed. In this state, the liquid ejection apparatus **11** is transported. When the transportation of the liquid ejection apparatus **11** is complete, the user turns on the power of the liquid ejection apparatus **11**. The control section **90** returns the liquid ejection apparatus **11** from the transport mode to the normal mode during a period from when the power is turned on to when the liquid consumption operation is performed for the first time. By returning to the normal mode, the supply flow paths **24** which were in the transport mode and in the closed state are opened. Accordingly, in the liquid ejection apparatus **11** set to the normal mode, both the supply flow paths **24** and the atmosphere communication flow paths **25** are in an open state. That is, the liquid in the liquid containers **23** can be supplied to the liquid ejection section **31**.

(75) Here, when the opening and closing mechanism is configured to be openable and closable by a manual operation, even though a user, a transporter, a repairman, or the like manually operates a manual operation section such as an operation lever from an open position to a closed position in order to transport the liquid ejection apparatus **11**, an operation of manually changing the manual operation section from the open position to the closed position is possible during transport. For this reason, during transportation of the liquid ejection apparatus **11**, a user, a transporter, a repairman,

or the like may erroneously change the manual operation section from the closed position to the open position, or may forget to return the change to the original closed position when the manual operation section is intentionally changed from the closed position to the open position. In these cases, the subsequent transport is performed in a state in which the supply flow paths **24** are opened. As a result, there is a risk that the ink will leak from the nozzles **33** of the liquid ejection section **31** during transportation of the liquid ejection apparatus **11**.
(76) In addition, when the opening and closing mechanism is configured to be openable and closable by a manual operation, there is the risk that a user will forget an operation of returning the manual operation section from the closed position to the open position even through transportation of the liquid ejection apparatus **11** is complete. In this case, there is a risk that the printing process is performed in a state in which the supply flow paths **24** are closed, ink is not ejected from the nozzle **33** of the liquid ejection section **31**, and an image is not printed on the medium M.
(77) The liquid ejection apparatus **11** of the present embodiment includes a configuration that suppresses a problem of a user or the like forgetting to change the manual operation section from the closed position to the open position after the transport is complete, and a printing process being performed in a state in which the supply flow paths **24** is closed. The liquid ejection apparatus **11** of the present embodiment does not include a manual operation section that switches the opening and closing state of the opening and closing mechanism **40**, but may include another manual operation section. In this case, under the power-off state, the manual operation section is locked to be inoperable by a lock mechanism (not shown), or the connection between the manual operation section and the opening and closing mechanism **40** is interrupted by a clutch (not shown) or the like, so that the opening and closing mechanism **40** is not switched even when the manual operation section is operated.
(78) Internal Configuration of Liquid Containers **23** and Configuration of the Maintenance Unit **50**
(79) Next, internal configuration of the liquid containers **23** and configuration of the maintenance unit **50** will be described with reference to FIG. **10**. As shown in FIG. **10**, the liquid containers **23** include the accommodation chamber **26** capable of containing liquid, the injection section **27** through which liquid is injected, an atmosphere communication section **64** for bringing the accommodation chamber **26** into communication with atmosphere, and a liquid outlet **65** for leading out the liquid in the accommodation chamber **26** to supply the liquid to the liquid ejection section **31**.
(80) The liquid containers **23** and the liquid ejection section **31** communicate with each other through the supply flow paths **24**. Liquid such as ink in the liquid containers **23** is supplied toward the liquid ejection section **31** via the supply flow paths **24** coupled to the liquid outlet **65**.
(81) When liquid such as ink in the liquid containers **23** is consumed and the liquid amount in the liquid containers **23** decreases, the pressure in the liquid containers **23** becomes lower than the atmospheric pressure. At this time, since the atmosphere communication flow paths **25** coupled to the atmosphere communication section **64** is in an open state, the atmosphere can flow into the accommodation chamber **26** from the atmosphere communication section **64**. Therefore, the pressure in the liquid containers **23** is maintained at atmospheric pressure.
(82) The liquid containers **23** include two injection flow paths **61** and **62** that bring the injection port **27**A of the injection section **27** into communication with the accommodation chamber **26**. The two injection flow paths **61** and **62** couple the injection port **27**A, which opens to the outside of the accommodation chamber **26**, and a delivery port **27**B, which opens to the inside of the accommodation chamber **26**. That is, the liquid containers **23** have the two injection flow paths **61** and **62**, whose upper end openings are the injection port **27**A and whose lower end openings are the delivery port **27**B. By coupling the supply port of the liquid bottle **75** to the injection port **27**A, the liquid can be injected from the liquid bottle **75** into the accommodation chamber **26** through the injection flow paths **61** and **62**. The delivery port **27**B is located at an upper limit position indicated by the upper limit mark UL (see FIG. **3**).

(83) The atmosphere communication section **64** communicates with the accommodation chamber **26** at a position above the highest liquid level, which is the height position of the delivery port **27**B. Further, the accommodation chamber **26** and the atmosphere communication section **64** may communicate with each other through an atmosphere communication passage (not shown) formed in a narrow and serpentine shape.
(84) As shown in FIG. **10**, since there are two the injection flow paths **61** and **62**, when the liquid is injected up to the delivery port **27**B at the time of liquid injection, the liquid injection automatically stops. Therefore, the user or the like can inject the liquid without checking the upper limit mark UL. At this time, the atmosphere communication flow path **25** is closed by the opening and closing mechanism **40**. Even when the atmosphere communication flow path **25** is closed, the liquid from the liquid bottle **75** is injected into the accommodation chamber **26** through one of the two injection flow paths **61** or **62**. At the same time, the air in the accommodation chamber **26** is supplied to the liquid bottle **75** through the other of the two injection flow paths **61** or **62**. As a result, the gas-liquid exchange is continuously performed between the liquid bottle **75** and the accommodation chamber **26** via two of the injection flow paths **61** and **62**, so that the liquid from the liquid bottle **75** replenishes the accommodation chamber **26**. When the liquid level L**1** in the accommodation chamber **26** reaches the height indicated by two dot chain line in FIG. **10** due to the injection of the liquid, the two delivery ports **27**B become blocked by liquid, so that gas-liquid exchange cannot be performed and injection of the liquid stops.
(85) Here, when the atmosphere communication flow path **25** is open and the inside of the accommodation chamber **26** communicates with atmosphere, even when the liquid level L**1** reaches the delivery port **27**B at the time of liquid injection, gas-liquid exchange is continued through the atmosphere communication flow path **25**, and thus the liquid injection does not stop. In order to avoid a situation in which the liquid injection does not stop, the control section **90** controls the opening and closing mechanism **40** to close at least the atmosphere communication flow path **25** when the control section **90** can predict that liquid injection will be performed.
(86) By closing the atmosphere communication flow path **25**, the delivery port **27**B will be blocked by the liquid reaching the height of the delivery port **27**B when the liquid level reaches the fill-up height at the time of liquid injection. As a result, since the gas-liquid exchange is stopped, the liquid injection from the liquid bottle **75** into the liquid container **23** automatically stops.
(87) In addition, the bottom wall of the accommodation chamber **26** is formed to be inclined in the depth direction Y so that the front surface side is higher. The liquid containers **23** each include the accommodation chamber **26**, a liquid outlet path **68** that communicates with the accommodation chamber **26** at the rear surface side, and the liquid outlet **65** that leads out the liquid that has passed via the liquid outlet path **68**. When liquid is consumed in the liquid ejection head **32**, the liquid stored in the accommodation chamber **26** is supplied to the liquid ejection section **31** via a filter and through the liquid outlet path **68**, the liquid outlet **65**, and the supply flow paths **24**.
(88) The supply flow paths **24** and the atmosphere communication flow paths **25** are configured to be openable and closable by the opening and closing mechanism **40** at respective intermediate positions. The supply flow paths **24** are opened and closed by the first opening and closing section **41**. The atmosphere communication flow path **25** is opened and closed by the second opening and closing section **42**. The first opening and closing section **41** and the second opening and closing section **42** are opened and closed by the drive of the motor **43**M of the drive mechanism **43**.
(89) In addition, as shown in FIG. **10**, the liquid ejection apparatus **11** includes a linear encoder **39** for detecting the position and the speed in the main scanning direction X of the reciprocating carriage **30**. The linear encoder **39** includes a linear code plate **39**B, which is provided on the housing **14**A and is parallel to the main scanning direction X, and an optical sensor **39**A, which is provided on the carriage **30**. A predetermined electric signal corresponding to the moving state of the carriage **30** is output from the optical sensor **39**A. An encoder signal including a digital pulse proportional to the amount of movement of the carriage **30** is output from the optical sensor **39**A as

an electrical signal. The encoder signal is input to the control section **90** (to be described later). The control section **90** performs position control and speed control of the carriage **30** based on the encoder signal. For example, the control section **90** performs position control to move the carriage **30** (liquid ejection section **31**) and stop the carriage **30** at the standby position HP and the exchange position EP.

(90) As shown in FIG. **10**, the maintenance unit **50** includes the cap **51** on an upper portion of a main body **52** in a state in which the cap **51** can be raised and lowered. The cap **51** is biased upward by a spring **54**. The cap **51** is configured to be movable up and down between a retracted position indicated by solid line in FIG. **10** and a raised position indicated by two dot chain line in FIG. **10**. When the cap **51** abuts the nozzle surface **32**A of the liquid ejection head **32**, a closed space communicating with the nozzles **33** is formed. Therefore, during non-printing, thickening and drying of the liquid such as ink in the nozzle **33** are suppressed. The maintenance unit **50** includes the pump **53** on a side portion of the main body **52**. When the pump **53** is driven by a motor (not shown) in a state in which the cap **51** abuts the nozzle surface **32**A to form the closed space, a negative pressure is introduced into the closed space, so that the liquid is forcibly discharged from the nozzles **33** into the cap **51**. The liquid (waste liquid) discharged into the cap **51** is collected in a waste liquid container (not shown) through a waste liquid tube (not shown) coupled to the cap **51**.

(91) The liquid ejection head **32** includes a pressure generation chamber (not shown) that communicates with the nozzles **33**, and a piezoelectric element (not shown) that changes the volume of the pressure generation chamber by, for example, an electrostrictive effect. The nozzles **33**, which open on the nozzle surface **32**A, communicate with the supply flow paths **24** via a flow path (not shown) in the liquid ejection section **31**. The piezoelectric element vibrates a vibration plate (not illustrated), which forms a part of the pressure generating chamber, to generate pressure fluctuation in the pressure generating chamber, and liquid such as ink is ejected from the nozzles **33** by using the pressure fluctuation. A liquid ejection method (for example, an inkjet method) of the liquid ejection head **32** is not limited to a piezoelectric method (piezo method), and may be, for example, a thermal method or a bubble method.

(92) Electrical Configuration of Liquid Ejection Apparatus **11**

(93) Next, with reference to FIG. **12**, an electric configuration of the liquid ejection apparatus **11** will be described.

(94) As shown in FIG. **12**, the control section **90** includes a central processing unit (CPU) **91** provided on a control board, a storage section **92**, and an interface unit (I/F) **93**.

(95) The I/F **93** performs data transmission and reception between an external host device **100** and the liquid ejection apparatus **11**. The host device **100** and the liquid ejection apparatus **11** may be directly coupled by a cable or the like, or may be indirectly coupled via a network or the like. In addition, data may be transmitted and received between the host device **100** and the liquid ejection apparatus **11** via wireless communication. The host device **100** is a terminal capable of giving an instruction for printing or scanning to the liquid ejection apparatus **11**. The host device **100** is a personal computer (PC), a tablet, a smartphone, a cellular phone, or the like.

(96) The CPU **91** is an arithmetic processing device that executes overall control of the liquid ejection apparatus **11**.

(97) The storage section **92** is a storage medium that secures an area for storing a program in which the CPU **91** operates, an operation area in which the program operates, and the like, and is composed of a storage element such as a RAM or an EPROM.

(98) The control section **90** creates print data based on image data received from the host device **100**, and controls the liquid ejection section **31**, the carriage motor **36**, the transport mechanism **70**, and the like based on the print data.

(99) The host device **100** may create print data, and the control section **90** may control the liquid ejection section **31**, the carriage motor **36**, the transport mechanism **70**, and the like based on the print data received from the host device **100**. Furthermore, a configuration may be adopted in

which the control section **90** creates print data based on an operation command input by a user using the operation section **15** of the operation panel **17**, and controls the liquid ejection section **31**, the carriage motor **36**, the transport mechanism **70**, and the like based on the print data.
(100) Further, the control section **90** drives a piezoelectric element provided in the liquid ejection section **31** to eject a liquid such as ink from the plurality of nozzles **33** toward the medium M. Further, the control section **90** supplies a drive signal to drive the carriage motor **36**.
(101) Here, a printing operation in which the liquid ejection apparatus **11** prints on the medium M will be described.
(102) The medium M accommodated in the medium accommodation section **18** is transported by the transport mechanism **70** from the upstream side to the downstream side in the transport direction −Y, which intersects with the main scanning direction X. The transport mechanism **70** transports the medium M in a state in which the transport mechanism **70** is supported by a support base (platen) (not illustrated) that is provided in a region facing the liquid ejection section **31** below the liquid ejection section **31**. Then, the liquid ejection section **31** ejects a liquid such as ink from the nozzles **33** onto the surface of the medium M supported by the support base which faces the liquid ejection section **31**. The liquid ejection section **31** reciprocates in the main scanning direction X while being mounted on the carriage **30**. Characters, images, or the like are printed on the medium M by alternately performing a printing operation of ejecting liquid from the nozzles **33** in a process in which the liquid ejection section **31** moves in the main scanning direction X and a transport operation of transporting the medium M to a next printing position. The medium M on which printing has been performed is discharged toward a medium discharge tray (not illustrated).
(103) The control section **90** receives an electric signal output from the linear encoder **39** (the optical sensor **39**A) for detecting the position and speed of the carriage **30**, which moves in accordance with drive of the carriage motor **36**.
(104) The control section **90** drives the transport mechanism **70** to move the medium M in a transport direction −Y, which intersects with the main scanning direction X.
(105) The control section **90** performs a maintenance operation on the liquid ejection section **31** by controlling the maintenance unit **50**.
(106) The control section **90** receives a command from the operation section **15** operated by a user, and performs various controls.
(107) The control section **90** uses an opening and closing detection section **72**, including an optical sensor or the like, to detect the open and closed states of the flow paths **24** and **25**, which are opened and closed by the opening and closing mechanism **40**. The opening and closing detection section **72** may be, for example, a rotary encoder or the like. The opening and closing detection section **72**, which is composed by a rotary encoder, detects the open and closed state of the opening and closing mechanism **40** by detecting the rotation angle of the cams **44** and **45**. The opening and closing detection section **72** of the present embodiment detects four switching positions as the open and closed states of the opening and closing mechanism **40**.
(108) The control section **90** calculates the position and the moving speed of the carriage **30** in the main scanning direction X using the electric signal output from the linear encoder **39** (the optical sensor **39**A). That is, the control section **90** controls the movement of the carriage **30**.
(109) The control section **90** moves the carriage **30** in the −X direction so that the carriage **30** is brought to abut with the sidewall of the housing **14**A. When the carriage **30** abuts the side wall of the housing **14**A, the movement of the carriage **30** in the −X direction is impeded, and the carriage **30** stops. When the movement of the carriage **30** in the −X direction is impeded, the driving load of the carriage motor **36** increases. The control section **90** defines the position of the carriage **30** when an increase in the drive load of the carriage motor **36** is detected as a reference position, and defines a standby position HP and an exchange position EP in the main scanning direction X.
(110) The operation section **15** and the display section **16** are electrically coupled to the control section **90**. The liquid ejection apparatus **11** may include a capacity detecting section (not shown)

for detecting the capacity of the liquid contained in each liquid container **23**. The capacity detecting section is, for example, a level sensor for detecting the capacity of the liquid contained in each liquid container **23**, and outputs a detection signal indicating the detection result to the control section **90**.
(111) The CPU **91** manages various controls including control of the liquid ejection apparatus **11** by executing a control program stored in the storage section **92**. The storage section **92** stores a control program that governs various controls in the liquid ejection apparatus **11** and reference data that is referred to in the control program. Various kinds of information for controlling the liquid ejection apparatus **11** by the control section **90** are stored in the storage section **92**. The storage section **92** according to the present embodiment stores various programs illustrated in flowcharts of FIGS. **14**, **15**, and **16** as examples of control programs. Specifically, a program (FIG. **14**) that includes control of the opening and closing mechanism **40** during the transport mode, a program (FIG. **15**) that includes control of replacement of the liquid ejection section **31**, and a program (FIG. **16**) that includes control of the opening and closing mechanism **40** during liquid injection are stored in the storage section **92**.
(112) The control section **90** (the CPU **91**) executes a program to execute control of the opening and closing mechanism **40** during the transportation mode (FIG. **14**), exchange control of the liquid ejection section **31** (FIG. **15**), and control of the opening and closing mechanism **40** during liquid injection (FIG. **16**).
(113) The control section **90** manages the mode of the liquid ejection apparatus **11**. When there is a command to transport the liquid ejection apparatus **11**, the second mode (transport mode) is selected, and when there is no command to transport the liquid ejection apparatus **11**, the first mode (normal mode) is selected. In this embodiment, the first mode is a normal mode when transportation of the liquid ejection apparatus **11** is not performed, and the second mode is a transportation mode when transportation of the liquid ejection apparatus **11** is performed.
(114) When the second mode (transport mode) is selected, at least one of the supply flow path **24** and the atmosphere communication flow path **25** are closed by the opening and closing mechanism **40**. When transportation of the liquid ejection apparatus **11** is complete and power is turned on the next time, the opening and closing mechanism **40** switches from the closed state to the open state during a period from the timing when power is next turned on to the timing of the start of the consumption of the liquid by the liquid ejection section **31**. The control section **90** detects that power was turned on when a user or the like operates the power operation section **15**A. Upon detection of power being turned on, the control section **90** checks the current mode, and if the second mode is selected, the control section **90** controls the opening and closing mechanism **40** to switch from the closed state to the open state until the first liquid consumption is started after power is turned on.
(115) For example, if the control section **90** is in the second mode at the time of power-on detection, the control section **90** may switch to the first mode at that time. Alternatively, when a liquid consumption operation is commanded after power-on detection, the control section **90** may switch from the second mode to the first mode before starting the liquid consumption operation. Here, the liquid consumption operation is a printing operation or a maintenance operation. The maintenance operation includes a cleaning operation and a flushing operation (dummy eject operation). Further, the liquid consumption operation includes a nozzle inspection performed by ejecting liquid from the nozzles **33**. For example, an inspection operation of ejecting liquid from the nozzles **33** in order to execute a nozzle check inspection for detecting nozzle clogging is also included. For example, when a first maintenance command is received after power-on is detected, the control section **90** switches from the second mode to the first mode before starting the maintenance operation. As a result, the control section **90** controls the opening and closing mechanism **40** to open the supply flow paths **24**.
(116) Accordingly, when a liquid consumption operation such as a liquid ejecting operation or a

liquid discharging operation is performed, liquid is supplied from the liquid containers **23** to the liquid ejection section **31** through the supply flow paths **24**. As a result, a required amount of liquid is ejected from the nozzles **33** of the liquid ejection section **31**. If the supply flowpath **24** is closed when liquid is consumed, there is a risk that an ejection failure or a discharge failure may occur in which a necessary amount of liquid is not ejected or discharged, but there is no such concern.
(117) Setting Screen
(118) Next, a setting screen **80** will be described with reference to FIG. **11**. FIG. **11** is the setting screen **80** displayed on the display section **16**. The user performs various settings on the liquid ejection apparatus **11** using the setting screen **80** by operating the operation section **15**. The setting screen **80** may be displayed on a display section (not shown) of the host device **100**, or various settings using the setting screen **80** may be performed on the liquid ejection apparatus **11** by operating an operation section (not shown) of the host device **100**.
(119) The liquid ejection apparatus **11** is configured to enable selection of a first mode in which, when power is turned off, power is turned off with the opening and closing mechanism **40** in the open state and a second mode in which, when power is turned off, power is turned off with the opening and closing mechanism **40** in the closed state.
(120) The setting screen **80** shown in FIG. **11** is an operation screen for selecting and setting one of the first mode or the second mode. Here, in the present embodiment, the first mode is the normal mode, and the second mode is the transport mode. The transport mode is a mode selected when the liquid ejection apparatus **11** is transported. A user, a transporter, a repairman, or the like selects the transport mode on the setting screen **80** when the liquid ejection apparatus **11** is to be transported. There is a possibility that liquid can leak from the nozzles **33** by the liquid ejection apparatus **11** being disposed during transportation at an inclination that exceeds a normal allowable inclination range or by, even if the inclination is within the allowable range, liquid moving to the liquid ejection section **31** side due to pressure fluctuation or the like acting on the liquid in the supply flow paths **24** due to shaking during transportation. The transport mode is a mode for suppressing leakage of the liquid from the nozzles **33** during this type of transport. Therefore, the second mode in which the supply flow paths **24** are closed is applied in the transport mode. The first mode in which the supply flow paths **24** are opened is applied in the normal mode other than the transport mode.
(121) As shown in FIG. **11**, the setting screen **80** is provided with a first selection section **81** for selecting the normal mode and a second selection section **82** for selecting the transport mode. Each selection section **81**, **82** is, for example, a selection button displayed on the screen, and is operated by the operation section **15**, a keyboard, a pointing device, or the like.
(122) As shown in FIG. **11**, when the transport mode is selected by operating the second selection section **82**, for example, the second selection section **82** displays an active selection state, and in conjunction with this, the first selection section **81** of the normal mode is displayed in, for example, a darkened non-selection state. On the other hand, when the first selection section **81** is operated to select the normal mode, the first selection section **81** displays, for example, an active selection state, and in association with this, the second selection section **82** of the transport mode is displayed in, for example, a darkened non-selection state. Instead of the configuration in which the selection is switched between the two selection sections **81** and **82**, a configuration in which the first mode (normal mode) and the second mode (transport mode) are selected by switching one selection section between the selected state and the unselected state may be employed.
(123) In addition, the setting screen **80** of the present embodiment is provided with a third selection section **83** that is selectively operated by a user or the like instructing that a liquid ejection section is to be replaced, and a fourth selection section **84** that is selectively operated by a user or the like when not instructing that a liquid ejection section be replaced. When the third selection section **83** is selected, the liquid ejection section exchange mode is set. When the liquid ejection section exchange mode is set, the “ON” indication of the third selection section **83** becomes active. When

the fourth selection section **84** is selected, the setting of the liquid ejection section exchange mode is canceled. When the liquid ejection section exchange mode is canceled, the “OFF” indication of the fourth selection section **84** becomes active. Instead of the configuration in which the selection is switched between the two selection sections **83** and **84**, a configuration in which one selection unit is switched between a selected state and a non-selected state may be employed.
(124) Further, the setting screen **80** is provided with a confirmation button **85** to be operated when confirming contents selected by the operation of each of the selection sections **81** to **84** are confirmed, and a cancel button **86** to be operated when the selected setting contents are to be canceled. When the confirmation button **85** is operated, the control section **90** confirms the setting contents selected on the setting screen **80** at that time.
(125) Opening and Closing Switching Control of the Opening and Closing Mechanism **40**
(126) Next, with reference to FIG. **13**, a description will be given of an opening and closing switching control content of the opening and closing mechanism **40**. The control section **90** controls the open and closed state of the opening and closing mechanism **40** by controlling the motor **43**M constituting the drive mechanism **43** of the opening and closing mechanism **40**.
(127) As shown in FIG. **13**, there are four opening and closing states of the opening and closing mechanism **40**, which are combinations of opening and closing of the first opening and closing section **41** and opening and closing of the second opening and closing section **42**. The control section **90** performs switching control to select one of the four switching positions by switching the rotational position of the cams **44** and **45** by the motor **43**M based on the detection signal of the opening and closing detection section **72**. That is, the control section **90** switches the cams **44** and **45** to the one of the four switching positions that corresponds to a command or the like at that time. In the present embodiment, each time their phase is switched by 90°, the cams **44** and **45**, which have a predetermined outer peripheral surface shape such as an elliptical shape, switch the combination of the open and closed states of the first opening and closing section **41** and the second opening and closing section **42** as illustrated in FIG. **13**.
(128) As shown in FIG. **13**, when the opening and closing mechanism **40** is at the first switching position, the first opening and closing section **41** opens and the second opening and closing section **42** opens. As a result, both the supply flow paths **24** and the atmosphere communication flow paths **25** are open. When the opening and closing mechanism **40** is at the second switching position, the first opening and closing section **41** close and the second opening and closing section **42** opens. As a result, the supply flow paths **24** are closed and the atmosphere communication flow paths **25** are open. When the opening and closing mechanism **40** is at the third switching position, the first opening and closing section **41** opens and the second opening and closing section **42** closes. As a result, the supply flow paths **24** are opened and the atmosphere communication flow paths **25** are closed. When the opening and closing mechanism **40** is in the fourth switching position, the first opening and closing section **41** is closed and the second opening and closing section **42** is closed. As a result, both the supply flow paths **24** and the atmosphere communication flow paths **25** are closed.
(129) The control section **90** selects the second mode in a case where there is a command to transport the liquid ejection apparatus **11**. In the second mode, when the power is turned off, the control section **90** turns off the power while the first opening and closing section **41** of the opening and closing mechanism **40** maintains the closed state.
(130) The control section **90** selects the first mode in a case where there is no command to transport the liquid ejection apparatus **11**. In the first mode, when the power is turned off, the control section **90** turns off the power while the first opening and closing section **41** of the opening and closing mechanism **40** maintains the open state.
(131) When there is a command to exchange the liquid ejection section **31**, the opening and closing mechanism **40** closes the supply flow paths **24**. In the present embodiment, when there is a command to exchange the liquid ejection section **31**, the control section **90** closes the supply flow

paths **24** by closing the first opening and closing section **41** of the opening and closing mechanism **40**. In this case, when there is a command to exchange the liquid ejection section **31**, the control section **90** controls the opening and closing mechanism **40** to close at least the supply flow paths **24**. That is, when there is a command to exchange the liquid ejection section **31**, only the supply flow paths **24** may be closed, or both the supply flow paths **24** and the atmosphere communication flow paths **25** may be closed. In the present embodiment, when there is a command to exchange the liquid ejection section **31**, the control section **90** controls the motor **43**M to switch the opening and closing mechanism **40** to the second switching position or the fourth switching position. At least as long as the supply flow paths **24** are closed, even if one end of the supply flow paths **24** is removed from the liquid ejection section **31** when the liquid ejection section **31** is exchanged, leakage of liquid from the one end of the supply flow paths **24** is suppressed.
(132) Alternatively, in a case where there is a command to exchange the liquid ejection section **31**, the control section **90** may control the opening and closing mechanism **40** to close at least the atmosphere communication flow paths **25**. That is, in a case where there is a command to exchange the liquid ejection section **31**, only the atmosphere communication flow paths **25** may be closed, or both the supply flow paths **24** and the atmosphere communication flow paths **25** may be closed. In the present embodiment, the control section **90** may switch the opening and closing mechanism **40** to the third switching position or the fourth switching position. When at least the atmosphere communication flow paths **25** are closed, even if one end of the supply flow paths **24** is detached from the liquid ejection section **31** when the liquid ejection section **31** is exchanged, the air chamber above the liquid surface in the accommodation chamber **26** is in a sealed state, and thus leakage of the liquid from the one end of the supply flow paths **24** is suppressed.
(133) When there is a command to exchange the liquid ejection section **31**, the control section **90** controls the carriage **30** to move the liquid ejection section **31** to the exchange position EP where the liquid ejection section **31** can be exchanged. When there is a command to turn off the power while the carriage **30** is at the exchange position EP, the control section **90** controls the carriage **30** to move the liquid ejection section **31** to the standby position HP.
(134) The control section **90** controls the opening and closing mechanism **40** to close at least the atmosphere communication flow paths **25** in accordance with the displacement of the cover **22** from the cover position to the exposed position recognized by the detection signal of the cover sensor **71**. That is, in accordance with the displacement of the cover **22** from the cover position to the exposed position, the opening and closing mechanism **40** may close only the atmosphere communication flow paths **25** or may close both the atmosphere communication flow paths **25** and the supply flow paths **24**. In the present embodiment, when the control section **90** detects that the cover **22** is displaced from the cover position to the exposed position based on the detection signal from the cover sensor **71**, the control section **90** controls the motor **43**M to switch the opening and closing mechanism **40** to the third switching position or the fourth switching position.
(135) The control section **90** manages various modes by flags stored in a predetermined storage area of the storage section **92**. The control section **90** has a flag for each type of mode, and manages the mode depending on whether the flag is “0” or “1”. The control section **90** manages whether the mode is the normal mode or the transport mode depending on whether the first flag is “0” or “1”. The control section **90** performs management based on whether the liquid ejection section exchange mode is on or off and whether the second flag is “0” or “1”. The control section **90** performs management based on whether the liquid injection mode is on or off and whether the third flag is “0” or “1”.
Action of Embodiment
(136) Next, the operation of the liquid ejection apparatus **11** according to the present embodiment will be described.
(137) During Normal Operation and During Transportation
(138) When transporting the liquid ejection apparatus **11**, the user selects the second mode, which

is the transport mode. Upon receiving the transportation command, the mode of the liquid ejection apparatus **11** shifts from the first mode, which is the normal mode, to the second mode, which is the transportation mode.
(139) Normally, the liquid ejection apparatus **11** is in the first mode, which is the normal mode. When transporting the liquid ejection apparatus **11**, a user, a transporter, a repairman, or the like causes the display section **16** to display the setting screen **80** illustrated in FIG. **11**. The setting screen **80** may be displayed on a display section of the host device **100** capable of communicating with the liquid ejection apparatus **11**. The user or the like operates the operation section **15** on the setting screen **80** to select the transport mode and then operates the confirmation button **85**. Then, the control section **90** accepts the transportation command. Upon receiving the transportation command, the control section **90** sets the second mode, which is the transportation mode. As a result, the liquid ejection apparatus **11** enters the second mode.
(140) Thereafter, when printing is finished in the normal mode, for example, the user operates the power operation section **15**A to turn off the power of the liquid ejection apparatus **11**. In order to transport the liquid ejection apparatus **11**, the user or the like who has set the transportation mode operates the power operation section **15**A to turn off the power of the liquid ejection apparatus **11**. The control section **90** executes a main routine shown in FIG. **14** at the timing when the power is turned on and off. With reference to FIG. **14**, the control contents in mainly the transport mode will be described below.
(141) First, in step S**11**, the control section **90** determines whether a power-on operation is detected. When a power-on operation is detected, the control section **90** proceeds to step S**12**, and when a power-on operation is not detected, the control section **90** ends this routine.
(142) In step S**12**, the control section **90** performs a power-on process. The power-on process is a process performed when the power is turned on, and includes, for example, an initialization process.
(143) In step S**13**, the control section **90** determines whether or not the mode is the transport mode. If it is the transport mode, the process proceeds to step S**14**, and if it is not the transport mode, the process proceeds to step S**16**. For example, when the transport mode is set during the power-on this time, the mode is the normal mode at the power-on time. Therefore, when the transport mode is to be set before the liquid ejection apparatus **11** is transported, the processes of steps S**14** and S**15** are not executed.
(144) Next, a user or the like, who is going to transport the liquid ejection apparatus **11**, selects the second selection section **82** on the setting screen **80** and then operates the confirmation button **85**. The control section **90** receives the confirmation operation in a state where the second selection section **82** is selected as a command for switching to the transport mode. After setting the transport mode, the user performs printing or the like if necessary and then subsequently, in order to transport the liquid ejection apparatus **11**, turns off the power by operating the power operation section **15**A.
(145) In step S**16**, the control section **90** determines whether or not a command to switch to the transport mode has been received. The control section **90** proceeds to step S**17** when a command to switch to the transport mode is received, and proceeds to step S**18** when no command to switch to the transport mode is received. Upon receiving a command to switch to the transport mode, the control section **90** proceeds to step S**17**.
(146) In step S**17**, the control section **90** sets the transport mode. For example, the control section **90** sets the first flag of the storage section **92** to "1".
(147) In step S**18**, the control section **90** determines whether a power-off operation is detected. When a power-off operation is detected, the process proceeds to step S**19**, and when a power-off operation is not detected, the process stands by until a power-off operation is detected. Note that the control section **90** executes necessary processes during this standby period, and executes the determination process of step S**18** regularly or irregularly, for example, by interrupt processing.

(148) In step S**19**, the control section **90** determines whether or not the mode is the transport mode. The control section **90** proceeds to step S**20** if it is in the transport mode, and proceeds to step S**21** if it is not in the transport mode. In this case, since the user or the like set the transport mode, the control section **90** proceeds to step S**20**.

(149) In step S**20**, the control section **90** closes the opening and closing mechanism **40**. Specifically, the control section **90** drives and controls the motor **43**M to switch the opening and closing mechanism **40** to the second switching position or the fourth switching position. As a result, at least the supply flow paths **24** are closed in the transport mode. For example, when the opening and closing mechanism **40** is switched to the second switching position, the first opening and closing section **41** closes and the second opening and closing section **42** opens, so that the supply flow paths **24** are closed and the atmosphere communication flow paths **25** are opened. Further, for example, when the opening and closing mechanism **40** is switched to the fourth switching position, the first opening and closing section **41** closes, and the second opening and closing section **42** closes, so that both the supply flow paths **24** and the atmosphere communication flow paths **25** are closed.

(150) In step S**21**, the control section **90** performs a power-off process. As the power-off process, the control section **90** performs, for example, a termination process of bringing the liquid ejection apparatus **11** into an appropriate termination state. Thereafter, a user, a transporter, a repairman, or the like transports the liquid ejection apparatus **11** in the power-off state. In the power-off state, in which the liquid ejection apparatus **11** is in the transport mode, the opening and closing mechanism **40** cannot be switched. For this reason, it is possible to suppress leakage of liquid from the nozzles **33**, which might occur if the opening and closing state of the opening and closing mechanism **40** were switched during transport of the liquid ejection apparatus **11** and the liquid ejection apparatus **11**, for example, were transported in a state in which the supply flow paths **24** were open.

(151) After transportation of the liquid ejection apparatus **11** is finished in this manner, the user or the like operates the power operation section **15**A to turn on the power of the liquid ejection apparatus **11**. The liquid ejection apparatus **11** at the time of power-on is in the state of the transport mode set during the previous power-on.

(152) In step S**11**, the control section **90** determines whether a power-on operation is detected. When the control section **90** detects a power-on operation, the process proceeds to step S**12**.

(153) In step S**12**, the control section **90** performs a power-on process. The control section **90** executes, for example, an initialization process as the power-on process.

(154) In step S**13**, the control section **90** determines whether or not the mode is the transport mode. Since the control section **90** determines that the mode is the transport mode, the process proceeds to step S**14**.

(155) In step S**14**, the control section **90** sets the normal mode. That is, the control section **90** switches from the transport mode to the normal mode.

(156) In the next step S**15**, the control section **90** opens the opening and closing mechanism **40**. Specifically, the control section **90** controls the motor **43**M to switch the opening and closing mechanism **40** to the first switching position. As a result, the first opening and closing section **41** of the opening and closing mechanism **40** opens the supply flow paths **24**, and the second opening and closing section **42** opens the atmosphere communication flow paths **25**.

(157) The above is the processing executed by the control section **90** from the timing at which the liquid ejection apparatus **11** is powered on to the timing at which the first consumption of liquid is started. That is, in a case where the transport mode (second mode) had been selected when power is turned on, the opening and closing mechanism **40** is switched from the closed state to the open state during a period from the timing when the power is turned on to the timing when the first consumption of the liquid by the liquid ejection section **31** is started. Specifically, the control section **90** controls the opening and closing mechanism **40** to open both the supply flow paths **24** and the atmosphere communication flow paths **25** during a period from a timing at which the power

is turned on to a timing at which consumption of the liquid by one of printing, cleaning, and flushing is started. The timing of opening the opening and closing mechanism **40** may be, for example, at the time that power is turned on or immediately before the liquid consumption operation is performed for the first time after the power was turned on.
(158) Exchange of Liquid Ejection Section
(159) The user, the repairman, or the like displays the setting screen **80** on the display section **16** and operates the operation section **15** to select the liquid ejection section exchange mode, thereby giving a liquid ejection section exchange command to the control section **90**. The setting screen **80** may be displayed on a display section of the host device **100**, and a liquid ejection section exchange command may be issued to the control section **90** by operating an operation section such as a keyboard of the host device **100** to select a liquid ejection section exchange mode. A user, a repairman, or the like selects the third selection section **83** on the setting screen **80** to select the liquid ejection section exchange mode. After the liquid ejection section exchange mode is selected, the user or the repairman operates the confirmation button **85**. The control section **90** receives this confirmation operation as an exchange command.
(160) Hereinafter, a liquid ejection section exchange control routine executed by the control section **90** will be described with reference to FIG. **15**.
(161) In step S**31**, the control section **90** determines whether an exchange command has been received. When a exchange command is received, this routine proceeds to step S**32**, and when a exchange command is not received, the routine ends.
(162) In step S**32**, the control section **90** controls the opening and closing mechanism **40** to close the supply flow paths **24**.
(163) In step S**33**, the control section **90** moves the liquid ejection section **31** to the exchange position EP.
(164) In step S**34**, the control section **90** determines whether an exchange end signal has been received. If an exchange end signal was not received, the process proceeds to step S**35**, and if an exchange end signal was received, the process proceeds to step S**36**.
(165) In step S**35**, the control section **90** determines whether a power-off operation is detected. If a power-off operation is detected, the process proceeds to step S**36**, and if a power-off operation is not detected, the process returns to step S**34**.
(166) In step S**36**, the control section **90** moves the liquid ejection section **31** to the standby position HP. Therefore, the control section **90** moves the carriage **30** to the standby position HP when the control section **90** receives the exchange end signal output when the user operates the operation section **15** at the end of exchange, or when the control section **90** detects a power-off operation without receiving the exchange end signal. Therefore, when the user or the repairman turns off the power during the exchange of the liquid ejection section, the carriage **30** moves from the exchange position EP to the standby position HP, so that the nozzle surface **32**A of the liquid ejection head **32** of the liquid ejection section **31** is capped with the cap **51**. As a result, a substantially closed space surrounded by the nozzle surface **32**A of the liquid ejection head **32** and the cap **51** is formed. As a result, drying of the liquid such as ink in the nozzle **33** is suppressed.
(167) During Liquid Injection
(168) At the time of liquid injection, the user opens the cover **13**A (scanner **13**) that covers the opening of the housing **14**A, and then further opens the cover **22** of the liquid storage units **20**.
(169) The control section **90** executes a cover opening operation interlocking control routine shown in FIG. **16** while the power of the liquid ejection apparatus **11** is turned on.
(170) In step S**41**, the control section **90** determines whether a cover opening operation is detected. When a cover opening operation for opening the cover **22** from the cover position to the exposed position is performed, the cover sensor **71** is switched from ON to OFF. When the cover sensor **71** is switched from ON to OFF, the control section **90** detects this as an opening operation of the cover **22**. The control section **90** proceeds to step S**42** when the cover opening operation is

detected, and proceeds to step S**43** when the cover opening operation is not detected.
(171) In step S**42**, the control section **90** controls the opening and closing mechanism **40** to close the atmosphere communication flow paths **25**. Specifically, the control section **90** closes at least the atmosphere communication flow paths **25** by switching the opening and closing mechanism **40** to the third switching position or the fourth switching position. For example, when the opening and closing mechanism **40** is switched to the third switching position, only the atmosphere communication flow paths **25** are closed. For example, when the opening and closing mechanism **40** is switched to the fourth switching position, both the supply flow paths **24** and the atmosphere communication flow paths **25** are closed.
(172) In this state, after opening the cover **22**, the user then opens the lid member **28** of the liquid container **23** to be replenished with liquid such as ink and exposes the injection port **27**A. By turning the liquid bottle **75** upside down and coupling the supply portion thereof to the injection port **27**A, a liquid such as ink is injected from the liquid bottle **75** into the accommodation chamber **26** via the injection port **27**A. At this time, before the liquid level L**1** of the liquid injected into the accommodation chamber **26** reaches the full level, gas-liquid exchange is performed between the liquid bottle **75** and the accommodation chamber **26** through the two injection flow paths **61** and **62**, so that injection of the liquid from the liquid bottle **75** into the accommodation chamber **26** continues.
(173) When the liquid level L**1** of the liquid injected into the accommodation chamber **26** reaches the full level, the delivery port **27**B is blocked by the liquid. Accordingly, gas-liquid exchange through the two injection flow paths **61** and **62** becomes impossible. As a result, the injection of the liquid from the liquid bottle **75** into the accommodation chamber **26** stops.
(174) Here, if the atmosphere communication flow paths **25** are not closed at the time of the liquid injection, since a state in which the inside of the accommodation chamber **26** communicates with atmosphere is maintained, the gas-liquid exchange would continue even if the liquid injected from the liquid bottle **75** into the accommodation chamber **26** exceeds the full level. Therefore, injection of the liquid from the liquid bottle **75** into the accommodation chamber **26** would continue. As a result, there is a risk that the liquid would overflow from the accommodation chamber **26**. On the other hand, in the present embodiment, when the cover **22** is opened, the control section **90** predicts liquid injection and drives the motor **43**M to switch the opening and closing mechanism **40** to the third switching position or the fourth switching position to close the atmosphere communication flow paths **25**, so that the inside of the accommodation chamber **26** does not communicate with atmosphere through the atmosphere communication flow paths **25**.
(175) As a result, when the liquid level L**1** in the accommodation chamber **26** reaches the full level at the time of liquid injection, the delivery port **27**B of the two injection flow paths **61** and **62** is blocked by the liquid, and thus the gas-liquid exchange stops. As a result, when the liquid level L**1** reaches the full level, the injection of the liquid from the liquid bottle **75** into the accommodation chamber **26** automatically stops.
(176) Therefore, the user can leave the accommodation chamber **26** until it is full without monitoring the amount of liquid supplied to the accommodation chamber **26**. Therefore, it is possible to avoid a situation in which the liquid overflows from the injection port **27**A and so is wastefully consumed or the liquid ejection apparatus **11** becomes contaminated with the liquid such as ink, which may occur in a case where the liquid is continuously supplied from the inside of the accommodation chamber **26** even when the liquid exceeds the full level.
(177) In step S**43**, the control section **90** determines whether a cover closing operation is detected. The control section **90** detects a cover closing operation when the cover sensor **71** is switched from ON to OFF. When a cover closing operation is detected, the control section **90** proceeds to step S**44**, and when a cover closing operation is not detected, the control section **90** stands by until a cover closing operation is detected.
(178) In step S**44**, the control section **90** controls the opening and closing mechanism **40** to open

the atmosphere communication flow paths **25**. The control section **90** switches the opening and closing mechanism **40** to, for example, the first switching position.

Effects of Embodiment

(179) Effects of the embodiment will be described. (1) The liquid ejection apparatus **11** includes the liquid ejection section **31** that performs printing by ejecting liquid from the nozzles **33**, the liquid containers **23** that include the accommodation chamber **26** and the injection port **27**A, the supply flow paths **24** that bring the liquid ejection section **31** into communication with the liquid containers **23**, and the opening and closing mechanism **40**. The opening and closing mechanism **40** is configured such that an open state, in which the supply flow paths **24** are open, and a closed state, in which the supply flow paths **24** are closed, can be switched between when the power is on and cannot be switched between when the power is off. The liquid ejection apparatus **11** is configured to enable selection of a first mode in which, when power is turned off, power is turned off with the opening and closing mechanism **40** in the open state and a second mode in which, when power is turned off, power is turned off with the opening and closing mechanism **40** in the closed state. According to this configuration, maintaining the closed state at the time of the power-off and maintaining the open state at the time of the power-off can be freely selected according to the situation, and it is possible to prevent the selected state from being changed intentionally or erroneously by a user, a transporter, a repairman, or the like at the time of the power-off. (2) In the liquid ejection apparatus **11**, the second mode is selected when there is a command to transport the liquid ejection apparatus **11**, and the first mode is selected when there is no command to transport the liquid ejection apparatus **11**. According to this configuration, since the second mode is selected when a command to transport is received, so it is possible to suppress leakage of liquid such as ink during transport. (3) In the liquid ejection apparatus **11**, when the second mode is selected, the opening and closing mechanism **40** switches from the closed state to the open state during the period from the next power-on timing to the start of consumption of the liquid by the liquid ejection section **31**. According to this configuration, it is possible to prevent forgetting to open the supply flow paths **24** after transportation. (4) The liquid ejection apparatus **11** includes the carriage **30** on which the liquid ejection section **31** is mounted and which is movable in the main scanning direction X. The liquid ejection section **31** is attachably and detachably provided with respect to the carriage **30**. One end of each supply flow path **24** is provided so as to be attachable to and detachable from the liquid ejection section **31**. When there is a command to exchange the liquid ejection section **31**, the opening and closing mechanism **40** closes the supply flow paths **24**. According to this configuration, it is possible to prevent the liquid in the supply flow path **24** from moving when the liquid ejection section **31** is removed. (5) The liquid containers **23** include the atmosphere communication flow paths **25** capable of communicating the inside of the accommodation chamber **26** with atmosphere. The opening and closing mechanism **40** is configured to open and close the atmosphere communication flow paths **25**. According to this configuration, a plurality of flow paths can be opened and closed by the single opening and closing mechanism **40**. (6) A state in which the opening and closing mechanism **40** closes only the supply flow paths **24** and a state in which the opening and closing mechanism **40** closes only the atmosphere communication flow paths **25** are selectable. According to this configuration, switching between opening and closing can be performed according to the situation. (7) The liquid ejection section **31** is attachably and detachably provided with respect to the carriage **30**. One end of each supply flow path **24** is provided so as to be attachable to and detachable from the liquid ejection section **31**. When there is a command to exchange the liquid ejection section **31**, the opening and closing mechanism **40** closes at least the supply flow paths **24**. According to this configuration, it is possible to prevent the liquid in the supply flow path **24** from moving when the liquid ejection section **31** is removed. (8) When there is a command to replace the liquid ejection section **31**, the opening and closing mechanism **40** closes at least the atmosphere communication flow paths **25**. According to this configuration, it is possible to prevent the liquid in the supply flow path **24** from

moving when the liquid ejection section **31** is removed. (9) The liquid ejection apparatus **11** further includes the carriage **30** that includes the liquid ejection section **31** and is movable in the main scanning direction X. When there is a command to replace the liquid ejection section **31**, the carriage **30** moves the liquid ejection section **31** to the exchange position EP where the liquid ejection section **31** can be exchanged. According to this configuration, it is possible to suppress the liquid ejection section **31** from being separated at an unexpected position. (10) The liquid ejection apparatus **11** further includes the cap **51** as an example of a closed space formation section that is capable of forming a closed space into which the nozzles **33** open, at the standby position HP at which the liquid ejection section **31** is capable of standing by. When there is a command to turn off the power while the carriage **30** is at the exchange position EP, the carriage **30** moves the liquid ejection section **31** to the standby position HP. According to this configuration, the evaporation of the liquid from the liquid ejection section **31** can be suppressed. (11) The liquid ejection apparatus **11** further includes the motor **43**M configured to drive the opening and closing mechanism **40**. The opening and closing mechanism **40** includes the first opening and closing section **41** that can open and close the supply flow paths **24**, and the first cam **44** that by drive of the motor **43**M switches opening and closing of the first opening and closing section **41**. According to this configuration, it is possible to easily switch between maintaining the closed state when the power is turned off and maintaining the open state when the power is turned off. (12) The opening and closing mechanism **40** includes the first opening and closing section **41** that can open and close the supply flow paths **24**, the second opening and closing section **42** that can open and close the atmosphere communication flow path **25**, the first cam **44** that by drive of the motor **43**M switches opening and closing of the first opening and closing section **41**, and the second cam **45** that by drive of the motor **43**M switches opening and closing of the second opening and closing section **42**. According to this configuration, it is possible to easily switch the plurality of opening and closing sections between the closed state being maintained at the time of power-off and the open state being maintained at the time of power-off. (13) The liquid ejection apparatus **11** further includes the cover **22** configured to be displaced between a cover position covering the liquid containers **23** and an exposed position exposing the liquid containers **23**. With the displacement of the cover **22** from the cover position to the exposed position, the opening and closing mechanism **40** closes at least the atmosphere communication flow paths **25**. According to this configuration, the liquid can be injected in a state in which the liquid containers **23** are not open to atmosphere.

Modifications

(180) The present embodiment can be modified as follows. The present embodiment and the following modifications can be implemented in combination with each other within a range that is not technically contradictory. As shown in FIG. **17**, the opening and closing mechanism **40** may include first opening and closing sections **41** for opening and closing the supply flow paths **24**, one for each of the liquid containers **23** provided on either side of the transport region FA in the main scanning direction X. The opening and closing mechanism **40** may include second opening and closing sections **42** for opening and closing the atmosphere communication flow paths **25**, one for each of the liquid containers **23** on either side of the transport region FA in the main scanning direction X. In the above-described embodiment, the merged flow path formed by merging the plurality of atmosphere communication flow paths **25** is opened and closed by the opening and closing mechanism **40**, but each of the atmosphere communication flow paths **25** may be individually opened and closed by the opening and closing mechanism **40**. In this case, as shown in FIG. **17**, since it is not necessary to route the atmosphere communication flow paths **25** to the position of a flow path holding section **77**, which bundles and holds all the supply flow paths **24** at a position in the housing **14**A close to the center in the main scanning direction X, it is possible to simplify the structure of the atmosphere communication mechanism. As shown in FIG. **18**, all the liquid containers **23** may be disposed on one side of the transport region FA in the main scanning direction X, and a common first opening and closing section **41** that opens and closes the supply

flow paths **24** may be provided for all the liquid containers **23**. In addition, a common second opening and closing section **42** that opens and closes the atmosphere communication flow paths **25** may be provided for all the liquid containers **23**. The opening and closing mechanism **40** may be configured to open and close only the supply flow paths **24**. In this case, the following operations (a) to (d) may be performed. (a) The supply flow path **24** is closed at the time of a transportation command, and the closed state is maintained even when the power is turned off. (b) The supply flow path **24** is closed at the time of a command to exchange the liquid ejection section **31**. (c) When there is a command to inject liquid, the supply flow path **24** is closed. (d) It is assumed that liquid is to be injected when the cover **22** is displaced from the cover position to the exposed position, and the supply flow path **24** is closed. When the opening and closing mechanism **40** is configured to be able to open and close the supply flow paths **24** and the atmosphere communication flow paths **25**, the following operations (1) to (11) may be performed. (1) Only the supply flow path **24** is closed when a transport command is received, and the closed state is maintained even at the time of power-off. (2) The supply flow path **24** and the atmosphere communication flow path **25** are closed when a transport command is received, and the closed state is maintained even at the time of power-off. (3) Only the supply flow path **24** is closed when a command to exchange the liquid ejection section **31** is issued. (4) Only the atmosphere communication flow path **25** is closed at the time of a command to exchange the liquid ejection section **31**. (5) Both the supply flow path **24** and the atmosphere communication flow path **25** are closed at the time of a command to exchange the liquid ejection section **31**. (6) When there is a command to inject liquid, only the supply flow path **24** is closed. (7) When there is a command to inject liquid, only the atmosphere communication flow path **25** is closed. (8) When there is a command to inject liquid, both the supply flow path **24** and the atmosphere communication flow path **25** are closed. (9) With the displacement of the cover **22** from the cover position to the exposed position, it is assumed that liquid is to be injected, and only the atmosphere communication flow path **25** is closed. (10) With the displacement of the cover **22** from the cover position to the exposed position, it is assumed that liquid is to be injected, and only the supply flow path **24** is closed. (11) With the displacement of the cover **22** from the cover position to the exposed position, it is assumed that liquid is to be injected, and both the supply flow path **24** and the atmosphere communication flow path **25** are closed. “Closing of the flow path in accordance with displacement of the cover” when a displacement detection section (for example, the cover sensor **71**) detects the displacement of the cover **22** is not limited to a configuration in which the flow path is closed by a separate drive mechanism (for example, the motor **43**M), and the flow path may be closed by the opening and closing mechanism **40** being linked with displacement of the cover **22**. At least the second opening and closing section **42** of the opening and closing mechanism **40** may be configured to operate in conjunction with movement of the cover **22**. For example, at least the atmosphere communication flow path **25** may be closed by operating the second cam **45** in conjunction with the movement of the cover **22** as the cover **22** is displaced from the cover position to the exposed position. Various commands may be executed from the operation section **15** (an operation panel, an operation button, or the like) included in the liquid ejection apparatus **11**, may be executed from various terminals (for example, the host device **100**), or may be executed from both. The opening and closing mechanism **40** may be provided in front of or behind the movement path of the carriage **30**. When the liquid containers **23** are provided on both sides of the transport path, the opening and closing mechanism **40** may be provided in a portion where the supply flow paths **24** and the atmosphere communication flow paths **25** are gathered in a single region. In this case, it is preferable that the single region is provided near the center in the width direction X, that is, at a position overlapping with the transport region FA in the vertical direction Z. In a case where the liquid containers **23** are provided on both sides of the transport region FA, the opening and closing mechanism **40** and the drive unit (for example, the motor **43**M) may be provided on both sides of the transport region FA, or only the opening and closing mechanism **40** may be provided on both sides of the transport region FA, and the opening and closing of the opening and closing mechanism **40** may be switched by transmitting drive from a single drive unit to the opening and closing mechanism **40** via the drive transmission section **43**A. When the opening and closing mechanism **40** includes the first opening and closing section **41** and the second opening and closing section **42**, the first opening and closing section **41** and the second opening and closing section **42** may be driven by separate drive mechanisms **43**. For example, the first cam **44** of the first opening and closing section **41** and the second cam **45** of the second opening and closing section **42** may be driven by separate motors **43**M. Although the first cam **44** of the first opening and closing section **41** and the second cam **45** of the second opening and closing section **42** are coaxially arranged, they may be arranged on separate axes parallel to each other or on separate axes intersecting each other. The motor **43**M may also serve as a motor related to the transport of the medium M, or may also serve as a motor related to the maintenance of the liquid ejection section **31**. In the case of the configuration in which the supply flow paths **24** are closed in the power-off state, the opening or closing state of the atmosphere communication flow paths **25** may be changed in conjunction with operation of a manual operation section or movement of the cover **22**. For example, gas-liquid exchange between the liquid bottle **75** and the accommodation chamber **26** may be automatically stopped when the liquid is injected into a liquid container **23** by closing the atmosphere communication flow path **25** which is in an open state in a power-off state by an operation of the manual operation section. According to this configuration, it is possible to suppress liquid leakage from the nozzles **33** caused by a pressure change accompanying the atmospheric pressure change in the accommodation chamber **26** of the liquid containers **23** in the power-off state, and it is possible to easily and appropriately perform the injection of the liquid into the liquid containers **23** in the power-off state. The opening and closing mechanism **40** may be driven by a drive section other than the motor **43**M. It is preferable that the drive mechanism **43** (a drive section such as the motor **43**M, the drive transmission section **43**A that transmits a driving force by a drive unit, and the like) of the opening and closing mechanism **40** is covered with a cover member or the like so as not to be exposed. According to this configuration, it is possible to prevent a user, a transporter, a repairman, or the like from forcibly opening and closing the opening and closing mechanism when the power is turned off, and it is possible to prevent the user, the transporter, the repairman, or the like from touching the opening and closing mechanism **40** while the opening and closing mechanism **40** is operating. When the liquid ejection apparatus **11** is powered on, the opening and closing mechanism **40** may be capable of opening and closing both in the state where the cover **22** is opened and in the state where the cover **22** is closed. The opening and closing mechanism **40** may be configured to be openable and closable in any state as long as the power is on. When the power of the liquid ejection apparatus **11** is on, the opening and closing mechanism **40** may be capable of opening and closing while the cover **22** is in a closed state, and may be incapable of opening and closing while the cover **22** is in an open state. According to this configuration, it is possible to suppress a concern that a user, a transporter, a repairman, or the like touches the opening and closing mechanism **40** during operation of the opening and closing mechanism **40**. The cover for switching the opened and closed state of the opening and closing mechanism **40** according to the displacement from the cover position to the exposed position is not limited to the cover **22** of the liquid storage units **20**, and may be the cover **13**A opened and closed with respect to the upper surface opening of the device main body **14**. That is, the open and closed state of the opening and closing mechanism **40** may be switched according to the displacement of the cover **13**A from the cover position to the exposed position. For example, the second opening and closing section **42** of the opening and closing mechanism **40** may switch the atmosphere communication flow paths **25** from the open state to the closed state in accordance with the displacement of the cover **13**A from the cover position to the exposed position. According to this configuration, when the cover **13**A (scanner **13**) is opened to expose the inside of the device main body **14** in order to exchange the liquid ejection section **31**, the atmosphere communication flow paths **25** are closed in

accordance with the displacement at this time of the cover **13**A from the cover position to the exposed position. Therefore, it is possible to suppress liquid leaking from the end section of the supply flow path **24** removed when the supply flow path **24** is removed from the liquid ejection section **31** at the joint section **35** at the time of exchanging the liquid ejection section **31**. In a case where the opening and closing mechanism **40** is capable of performing both opening and closing by the drive mechanism **43** and manual opening and closing, switching between opening and closing may be disabled by providing a configuration that locks the operation of the manual operation section when the power is off. The opening and closing detection section **72** may not be provided. An operation for switching the opening and closing mechanism **40** to the open state may be performed each time the power is turned on or the consumption of liquid by the liquid ejection section **31** is started after the power is turned on. Accordingly, it is possible to suppress the occurrence of a switching failure of the opening and closing mechanism **40** to the open state in a case where the opening and closing detection section **72** is not provided, a case where the opening and closing detection section **72** fails, a case where the control section **90** fails and it becomes unclear that the mode is the transport mode, or the like. The liquid ejection apparatus **11** is not limited to an inkjet printer that discharges a liquid such as ink to a medium M such as paper, and may be a textile printing apparatus that discharges a liquid such as ink to a fabric. The liquid ejection apparatus **11** is not limited to a serial printer, and may be a line printer or a page printer.
(181) The liquid ejection apparatus **11** may be a printer having only a printing function without including the scanner **13** (image reading unit).
(182) Hereinafter, technical ideas derived from the above-described embodiments and modifications and effects thereof will be described. (A) A liquid ejection apparatus includes a liquid ejection section configured to print by discharging a liquid from a nozzle, a liquid container having an accommodation chamber that stores liquid and an injection port that communicates with the accommodation chamber and into which the liquid is configured to be injectable from the outside, a supply flow path communicating between the liquid ejection section and the liquid container, and an opening and closing mechanism that is configured to switch such that an open state in which the supply flow path is opened and a closed state in which the supply flow path is closed when power is turned on and not to switch when power is turned off. The liquid ejection apparatus is configured to enable selection of a first mode in which, when power is turned off, power is turned off with the opening and closing mechanism in the open state and a second mode in which, when power is turned off, power is turned off with the opening and closing mechanism in the closed state.
(183) According to this configuration, maintaining the closed state at the time of the power-off and maintaining the open state at the time of the power-off can be freely selected according to the situation, and it is possible to prevent the selected state from being changed intentionally or erroneously by a user, a transporter, a repairman, or the like at the time of the power-off. (B) In the liquid ejection apparatus, when there is a command to transport the liquid ejection apparatus, the second mode may be selected and when there is no command to transport the liquid ejection apparatus, the first mode may be selected.
(184) According to this configuration, since the second mode is selected when a command to transport is received, so it is possible to suppress leakage of liquid such as ink during transport. (C) In the liquid ejection apparatus, when the second mode is selected, the opening and closing mechanism may switch from the closed state to the open state during the period from the next power-on timing to the start of consumption of the liquid by the liquid ejection section.
(185) According to this configuration, it is possible to prevent forgetting to open the supply flow paths **24** after transportation. (D) The liquid ejection apparatus may further include a carriage that has the liquid ejection section and is movable in a main scanning direction, wherein the liquid ejection section is detachably attached to the carriage, one end of the supply flow path is detachably attached to the liquid ejection section, and the opening and closing mechanism closes

the supply flow path when a command to exchange the liquid ejection section is issued. According to this configuration, it is possible to prevent the liquid in the supply flow path from moving when the liquid ejection section is separated. (E) In the liquid ejection apparatus, the liquid container may include an atmosphere communication flow path capable of communicating the inside of the accommodation chamber with atmosphere, and the opening and closing mechanism may be configured to be capable of opening and closing the atmosphere communication flow path.
(186) According to this configuration, a plurality of flow paths can be opened and closed by one opening and closing mechanism. (F) In the liquid ejection apparatus, a state in which the opening and closing mechanism closes only the supply flow path and a state in which the opening and closing mechanism closes only the atmosphere communication flow path can be selected.
(187) According to this configuration, switching between opening and closing can be performed according to the situation. (G) The liquid ejection apparatus further includes a carriage on which the liquid ejection section is mounted and which is movable in the main scanning direction, wherein the liquid ejection section is detachably attached with respect to the carriage, and wherein one end of the supply flow path is detachably attached with respect to the liquid ejection section.
(188) When there is a command to replace the liquid ejection section, the opening and closing mechanism may close at least the supply flow path.
(189) According to this configuration, it is possible to prevent the liquid in the supply flow path from moving when the liquid ejection section is separated. (H) The liquid ejection apparatus further includes a carriage mounted with the liquid ejection section and movable in the main scanning direction, wherein the liquid ejection section is detachably attached with respect to the carriage, and wherein one end of the supply flow path is detachably attached with respect to the liquid ejection section.
(190) The opening and closing mechanism may close at least the atmosphere communication flow path when there is a command to replace the liquid ejection section.
(191) According to this configuration, it is possible to prevent the liquid in the supply flowpath from moving when the liquid ejection section is separated. (I) The liquid ejection apparatus may further include a carriage that has the liquid ejection section and is movable in a main scanning direction, and the carriage may move the liquid ejection section to an exchange position at which the liquid ejection section is exchangeable when a command to exchange the liquid ejection section is issued.
(192) According to this configuration, it is possible to suppress the liquid ejection section from being separated at an unexpected position. (J) The liquid ejection apparatus may further include a closed space formation section that is capable of forming a closed space in which the nozzle opens, at a standby position at which the liquid ejection section is capable of standing by, and the carriage may move the liquid ejection section to the standby position when there is a command to turn off power while the carriage is at the exchange position.
(193) According to this configuration, evaporation of the liquid from the liquid ejection section can be suppressed. (K) The liquid ejection apparatus may further include a motor configured to drive the opening and closing mechanism, and the opening and closing mechanism may have a first opening and closing section configured to open and close the supply flow path and a first cam configured to switch opening and closing of the first opening and closing section by the motor drive.
(194) According to this configuration, it is possible to easily switch between maintaining the closed state when the power is turned off and maintaining the open state when the power is turned off. (L) The liquid ejection apparatus may further include a motor configured to drive the opening and closing mechanism, the opening and closing mechanism has a first opening and closing section configured to open and close the supply flow path, a second opening and closing section configured to open and close the atmosphere communication flow path, a first cam configured to switch between opening and closing of the first opening and closing section by driving of the motor, and a

second cam configured to switch between opening and closing of the second opening and closing section by driving of the motor.
(195) According to this configuration, it is possible to easily switch the plurality of opening and closing sections between the closed state being maintained at the time of power-off and the open state being maintained at the time of power-off. (M) The liquid ejection apparatus may further include a cover configured to be displaced between a cover position covering the liquid container and an exposed position exposing the liquid container, and the opening and closing mechanism may close at least the atmosphere communication flow path in response to displacement of the cover from the cover position to the exposed position.
(196) According to this configuration, it is possible to inject the liquid in a state where the liquid container is not open to the atmosphere.

Claims

1. A liquid ejection apparatus comprising: a liquid ejection section configured to print by ejecting a liquid from a nozzle; a liquid container including an accommodation chamber configured to store liquid and an injection port that communicates with the accommodation chamber and that is configured to be injected with liquid from outside; a supply flow path communicating between the liquid ejection section and the liquid container; and an opening and closing mechanism configured to, when power is turned on, enable, and, when power is turned off, disable switching between an open state, in which the supply flow path is open, and a closed state, in which the supply flow path is closed, wherein the liquid ejection apparatus is configured to enable selection of a first mode in which, when power is turned off, power is turned off with the opening and closing mechanism in the open state, and a second mode in which, when power is turned off, power is turned off with the opening and closing mechanism in the closed state, when there is a command to transport the liquid ejection apparatus, the second mode is selected, and when there is no command to transport the liquid ejection apparatus, the first mode is selected.
2. The liquid ejection apparatus according to claim 1, wherein when the second mode is selected, the opening and closing mechanism switches from the closed state to the open state in a period from when power is next turned on to when consumption of the liquid by the liquid ejection section is started.
3. The liquid ejection apparatus according to claim 1, further comprising: a carriage on which the liquid ejection section is mounted and that is configured to move in a main scanning direction, wherein the liquid ejection section is attachable to and detachable from the carriage, the supply flow path is provided such that one end is attachable to and detachable from the liquid ejection section, and when there is a command to replace the liquid ejection section, the opening and closing mechanism closes the supply flow path.
4. The liquid ejection apparatus according to claim 1, further comprising: a motor configured to drive the opening and closing mechanism, wherein the opening and closing mechanism includes a first opening and closing section configured to open and close the supply flow path and a first cam configured to switch an opening state and a closing state of the first opening and closing section by drive of the motor.
5. The liquid ejection apparatus according to claim 1, wherein the liquid container has an atmosphere communication flow path configured to communicate inside of the accommodation chamber with atmosphere and the opening and closing mechanism is configured to open and close the atmosphere communication flow path.
6. The liquid ejection apparatus according to claim 5, configured to enable selection of a state in which the opening and closing mechanism closes only the supply flow path and a state in which the opening and closing mechanism closes only the atmosphere communication flow path.
7. The liquid ejection apparatus according to claim 5, further comprising: a motor configured to

drive the opening and closing mechanism, wherein the opening and closing mechanism includes a first opening and closing section configured to open and close the supply flow path, a second opening and closing section configured to open and close the atmosphere communication flow path, a first cam configured to switch the opening state and the closing state of the first opening and closing section by drive of the motor, and a second cam configured to switch the opening state and the closing state of the second opening and closing section by drive of the motor.

8. The liquid ejection apparatus according to claim 5, further comprising: a cover configured to be displaced between a cover position at which the liquid container is covered and an exposed position at which the liquid container is exposed, wherein when the cover is displaced from the cover position to the exposed position, the opening and closing mechanism closes at least the atmosphere communication flow path.

9. The liquid ejection apparatus according to claim 5, further comprising: a carriage on which the liquid ejection section is mounted and that is configured to move in a main scanning direction, wherein the liquid ejection section is attachable to and detachable from the carriage, the supply flow path is provided such that one end is attachable to and detachable from the liquid ejection section, and when there is a command to replace the liquid ejection section, the opening and closing mechanism closes at least the atmosphere communication flow path.

10. The liquid ejection apparatus according to claim 9, wherein when there is a command to replace the liquid ejection section, the carriage moves the liquid ejection section to an exchange position where the liquid ejection section is replaceable.

11. The liquid ejection apparatus according to claim 10, further comprising: a closed space formation section configured to form, at a standby position where the liquid ejection section is configured to stand by, a closed space into which the nozzle opens, wherein when there is a command to turn off power when the carriage is at the exchange position, the carriage moves the liquid ejection section to the standby position.

12. The liquid ejection apparatus according to claim 5, further comprising: a carriage on which the liquid ejection section is mounted and that is configured to move in a main scanning direction, wherein the liquid ejection section is attachable to and detachable from the carriage, the supply flow path is provided such that one end is attachable to and detachable from the liquid ejection section, and when there is a command to replace the liquid ejection section, the opening and closing mechanism closes at least the supply flow path.

13. The liquid ejection apparatus according to claim 12, wherein when there is a command to replace the liquid ejection section, the carriage moves the liquid ejection section to an exchange position where the liquid ejection section is replaceable.

14. The liquid ejection apparatus according to claim 13, further comprising: a closed space formation section configured to form, at a standby position where the liquid ejection section is configured to stand by, a closed space into which the nozzle opens, wherein when there is a command to turn off power when the carriage is at the exchange position, the carriage moves the liquid ejection section to the standby position.